

NHS Data Quality Analysis

Project Overview

This project analyses the quality of a synthetic NHS patient dataset containing **10,000 patient records**.

The project uses **Python, SQL and Power BI** to identify, measure, clean and communicate data-quality issues. The aim is to demonstrate how data-quality checks can be applied throughout the data-analysis process, from initial data cleaning through to SQL analysis and dashboard reporting.

Objectives

- Assess data completeness
- Validate NHS Numbers
- Validate Patient IDs
- Check date consistency
- Identify duplicate records
- Validate categorical fields
- Identify missing values
- Calculate an overall Data Quality Score
- Clean and prepare the dataset
- Analyse the cleaned dataset using SQL
- Create an interactive Power BI dashboard
- Present key findings and recommendations clearly

Tools & Technologies

- **Python**
- **Pandas**
- **MySQL**
- **SQL**
- **Power BI**
- **Microsoft Excel / CSV**
- **GitHub**

Data Quality Results

Data Quality Dimension	Score
Completeness	93.85%

Data Quality Dimension	Score
Validity	99.18%
Uniqueness	99.91%
Consistency	95.09%
Overall Data Quality	97.01%

Key Data Quality Issues

The initial data-quality assessment identified several issues:

- **397** missing NHS Numbers
- **2,065** missing Gender values
- **1,291** missing Postcodes
- **1,722** missing GP Practice values
- **1,292** missing Diagnosis Codes
- **491** invalid admission/discharge date relationships
- **9** duplicate NHS Number occurrences

These issues were identified and flagged before the cleaned dataset was used for further analysis.

Python Data Cleaning

Python and Pandas were used as the first stage of the data-quality process.

The analysis included:

- Checking for missing values
- Validating NHS Numbers and Patient IDs
- Checking duplicate records
- Validating admission and discharge dates
- Checking categorical fields
- Creating data-quality flags
- Calculating data-quality measures
- Preparing the cleaned dataset for SQL analysis

This provided a structured and repeatable approach to identifying data-quality problems.

SQL Analysis

The cleaned dataset was successfully imported into **MySQL** for further analysis.

SQL was used to explore the cleaned data and answer business-style questions around patient records and data quality.

The SQL analysis included:

- Total number of patient records
- Missing and invalid data analysis
- Duplicate record analysis
- Department-level analysis
- Gender distribution
- Patient record summaries
- Date-related analysis
- Data-quality issue counts
- Comparison of valid and invalid records

The completed SQL work demonstrates how the cleaned dataset can be queried and analysed using relational database techniques.

Power BI Dashboard

The final **Power BI dashboard** was created to communicate the data-quality results in an interactive and easy-to-understand format.

The dashboard includes key performance indicators and visualisations covering:

- Total patient records
- Overall Data Quality Score
- Valid and invalid records
- Missing data
- Data-quality issues by field
- Gender data quality
- Date-quality checks
- Patient ID and NHS Number checks
- Department-level analysis
- Data-quality issue breakdowns

Interactive slicers and visualisations allow users to explore the dataset and identify areas requiring attention.

Key Insights

The analysis showed that the overall quality of the dataset was **97.01%**, indicating a generally high level of data quality.

However, completeness was the main area requiring improvement. Missing Gender, GP Practice and Diagnosis Code values accounted for a significant proportion of the identified issues.

The analysis also identified **491 records with inconsistent admission and discharge dates**, highlighting the importance of date validation in healthcare datasets.

Although the uniqueness score was very high at **99.91%**, duplicate NHS Number occurrences were still identified and required investigation.

The SQL analysis provided further insight into the distribution of records and data-quality issues, while the Power BI dashboard made these findings easier to monitor and communicate.

Recommendations

Based on the findings, the following improvements are recommended:

1. **Improve data-entry controls** to reduce missing Gender, GP Practice and Diagnosis Code values.
2. **Introduce validation rules** for NHS Numbers and other important identifiers.
3. **Apply date validation checks** to prevent discharge dates being recorded before admission dates.
4. **Monitor duplicate NHS Numbers** to reduce the risk of duplicate patient records.
5. **Use mandatory fields** for critical patient information where appropriate.
6. **Carry out regular data-quality checks** rather than waiting until the data is required for reporting.
7. **Use Power BI dashboards** to monitor data-quality performance and identify recurring problems.
8. **Review data-quality scores regularly** to measure whether improvements are having an impact.

Project Structure

The project is organised into separate stages:

NHS-Data-Quality-Analysis/

```
|
|
| └─ data/
|   └─ raw/
|   └─ cleaned/
|
|
| └─ python/
|   └─ data_quality_analysis.py
|
```

```
├─ sql/
|   ├── data_quality_analysis.sql
|   └─ README.md
|
├─ powerbi/
|   └─ NHS_Data_Quality_Dashboard.pbix
|
├─ reports/
|   └─ NHS_Data_Quality_Analysis_Report.pdf
|
└─ README.md
```

The **Python** section contains the data-cleaning and quality-checking work. The **SQL** section contains the completed MySQL analysis. The **Power BI** section contains the interactive dashboard used to present the findings, while the **reports** section contains the project documentation.

End-to-End Data Analysis Process

The project follows a complete data-analysis workflow:

Raw Dataset → Python Data Cleaning → Data Quality Checks → Cleaned Dataset → MySQL/SQL Analysis → Power BI Dashboard → Insights & Recommendations

This demonstrates practical experience across the full data-analysis lifecycle rather than focusing on dashboard development alone.

Disclaimer

This project uses **synthetic/anonymised data for educational and portfolio purposes**. It does not contain real NHS patient information.